

Food Ordering Customer Behavior: Tableau Analytics Story

Project Description:

The Food Ordering Customer Story is a Tableau-based analytics project that helps stakeholders understand customer ordering behavior across cities, cuisines, meal types, and restaurant categories. The workbook analyzes a food-delivery behavior dataset of 50,000 orders placed by 4,000 unique customers across six Indian cities — Mumbai, Pune, Bangalore, Delhi, Chandigarh, and Hyderabad.

The workbook is built using Tableau Desktop. The source CSV (food_ordering_behavior_dataset.csv) is materialized as a Tableau Extract (.hyper file) embedded inside a packaged workbook (.twbx), making it fully self-contained and portable. Twelve worksheets, three input parameters (City, Cuisine, Meal Type), and two calculated fields (Age Group and a delivery-time bin) are combined into a single operational dashboard and a nine-point guided story.

The workbook provides an easy-to-use, interactive interface where stakeholders can filter by city, cuisine, or meal type and instantly see how KPIs, revenue, ratings, and delivery-time patterns respond. This helps evaluators and business teams review ordering behavior at a glance and supports data-driven decisions around menu mix, city-level operations, and delivery performance.

Scenarios

Scenario 1:

A stakeholder wants to review ordering behavior for a single city, cuisine, and meal occasion:

- City Parameter: Delhi
- Cuisine Parameter: Biryani
- Meal Type Parameter: Lunch

Every worksheet that reads the parameter-driven filter fields (City / Cuisine / Meal Type Parameter calc) instantly narrows to **Delhi** → **Biryani** → **Lunch** orders only, letting the stakeholder inspect that specific slice of the business without touching the underlying data.

Scenario 2:

A stakeholder wants the full, unfiltered picture across all cities, cuisines, and meal types:

- City Parameter: All
- Cuisine Parameter: All
- Meal Type Parameter: All

With every parameter set to “**All**”, the boolean helper calculations return TRUE for every row, and the dashboard displays the full aggregate view: 50,000 total orders, Rs. 27,386,555 in total order value, Rs. 2,982,136 in total delivery fees, and an average customer rating of 2.99 out of 5.

Scenario 3:

A stakeholder wants to understand generational preferences without using any parameter:

- Dimension used: Age Group (calculated field)
- Segments: Teen (<20), Gen Z (20–30), Millennial (31–40), Adults (41+)

The Age Group calculated field automatically buckets every customer by age and feeds the grouped bar chart. The system shows that restaurant-type preference (Premium: 16,859 orders; Budget: 16,660; Mid-range: 16,481) is spread evenly across all four age groups, indicating restaurant-category choice is **not strongly age-dependent** in this dataset.

Technical Architecture

Description

The Food Ordering Customer Story follows a **three-layer architecture**.

The **Data Layer** consists of the raw CSV file (food_ordering_behavior_dataset.csv), which is connected in Tableau and materialized as a fast, portable Extract (.hyper) embedded inside the packaged workbook.

The **Interactivity Layer** is built from calculated fields and parameters. Age Group segments customers by generation; the City, Cuisine, and Meal Type Parameters (with their boolean helper calculations) let a viewer filter the entire workbook to a single slice of data.

The **Visualization Layer** contains twelve purpose-built worksheets covering bar, grouped bar, pie, donut, heat-map, and KPI text-table chart types, consolidated into Dashboard 1 (a combined operational overview) and Story 1 (a nine-point guided narrative for stakeholder presentation).

Architecture Flow



Figure 1. End-to-end workflow from source data to the interactive story.

Pre-requisites

Before opening and exploring the Food Ordering Customer Story, ensure that the following software, files, and concepts are available and understood:

- **Tableau Desktop** – Used to author, open, and edit the workbook, its worksheets, dashboard, and story.
- **Tableau Public or Tableau Reader** – Used as free alternatives to view and interact with the packaged workbook (read-only for Reader).
- **food_ordering_behavior_dataset.csv** – The 50,000-record source dataset the workbook is built from.
- **Tableau Extract / .hyper engine** – Used to materialize the CSV into a fast, offline-ready extract embedded in the .twbx file.
- **Calculated Fields & Parameters knowledge** – Basic understanding of how Tableau parameters and boolean calculated fields drive interactive filtering.
- **Data Visualization Concepts** – Familiarity with bar, grouped bar, pie, donut, heat-map, and KPI text-table chart types and when to use each.
- **Tableau Viz Extension** – Used to render the custom donut chart on the Meal Type vs Customer Rating worksheet.
- **Web Browser (Google Chrome/Microsoft Edge)** – Used to view the workbook if published to Tableau Public.
- **Windows/Mac Operating System** – Supported platforms for Tableau Desktop, Public, and Reader.

Required Tools & Setup

Unlike a coded application, this project has no package dependencies to install — it only requires the Tableau viewing/authoring tool itself plus the workbook and dataset files:

1. Install Tableau Desktop, or Tableau Public (free), or Tableau Reader (free, read-only)
2. Place `food_ordering_behavior_dataset.csv` in the project folder
3. Open `Food Ordering Customer Story.twbx` directly — the embedded extract unpacks automatically

Milestone 1: Project Setup

Step 1: Project Structure

Content:

The Food Ordering Customer Story project was created using Tableau Desktop. The project contains the packaged workbook, the source dataset, and this documentation file. The structure is organized to keep the data source, visual assets, and reference documentation separate:

```
Food_Ordering_Customer_Story/  
  food_ordering_customer_story.twbx (packaged workbook: worksheets, dashboard, story,  
  embedded extract)  
  food_ordering_behavior_dataset.csv (source dataset, 50,000 records)  
  Food_Ordering_Customer_Story_Documentation.docx (this documentation / README)
```

Step 2: Create Data Source Connection

In Tableau Desktop, connect to the CSV file (`food_ordering_behavior_dataset.csv`) as a text file data source, then convert the live connection to an Extract (Data → Extract Data) so the workbook can be packaged and shared without requiring the original CSV on the viewer's machine.

Step 3: Install Required Tools

Command / Action:

```
Install Tableau Desktop (trial/licensed) or Tableau Public (free) from tableau.com
```

Milestone 2: Data Modeling

Step 1: Dataset

Content:

The dataset contains 50,000 order-level records with fields such as `city`, `cuisine`, `meal_type`, `restaurant_type`, `order_value`, `delivery_fee`, `time_taken_to_order`, `rating_given`, `age`, `mood`, `hunger_level`, `company`, and `rainy_weather`. This dataset is used to build every worksheet in the workbook.

Step 2: Calculated Field & Parameter Building

Content:

Calculated fields and parameters are layered on top of the raw columns to enable segmentation and interactivity. The Age Group field buckets customers into four generational segments, and three parameters (City, Cuisine, Meal Type) power the interactive filtering used throughout the dashboard and story.

Formula (Age Group):

```
IF [age] < 20 THEN "Teen"
ELSEIF [age] <= 30 THEN "Gen Z"
ELSEIF [age] <= 40 THEN "Millennial"
ELSE "Adults"
END
```

Milestone 3: Worksheet & Dashboard Development

Step 1: Worksheet Development

Content:

Each worksheet isolates one analytical question and uses an appropriate Tableau chart type (bar, grouped bar, pie, donut, heat map, or KPI text table). Worksheets are built by dragging dimensions and measures onto the Rows, Columns, Color, and Label shelves.

Build Steps (example: Top 5 Cities by Order Volume)

```
Drag City to the Rows shelf and Order Id (as Count) to the Columns shelf / Label.  
Sort descending by the count measure, then apply a Top N filter  
(Filter → City → Top tab → By Field → Top 5 → Count of Order Id).  
Set the mark type to Bar, add the count measure to Color and Label, and add the chart title.
```

Step 2: Open & Run the Workbook

1. Download or copy “food Ordering Customer Story.twbx” to a local folder.
2. Double-click the file, or open it from within Tableau Desktop via File → Open.
3. Tableau unpacks the embedded extract automatically – no manual extraction is needed.
4. Confirm the data source shows 50,000 rows (Data pane → right-click data source → View Data).

Milestone 4: Dashboard & Story Interface

Dashboard 1 — Combined Overview

Description:

Dashboard 1 arranges eight worksheets plus the KPI summary onto a single canvas, giving a one-screen operational overview of ordering behavior: headline KPIs, revenue by meal type, city leaderboard, delivery-time distribution, rating by meal type, age-group vs restaurant-type breakdown, city-vs-cuisine heat map, and cuisine share.

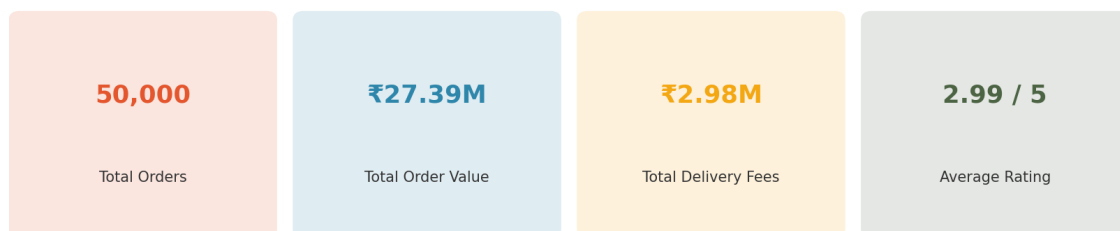


Figure 2. Headline KPI summary computed from all 50,000 orders.

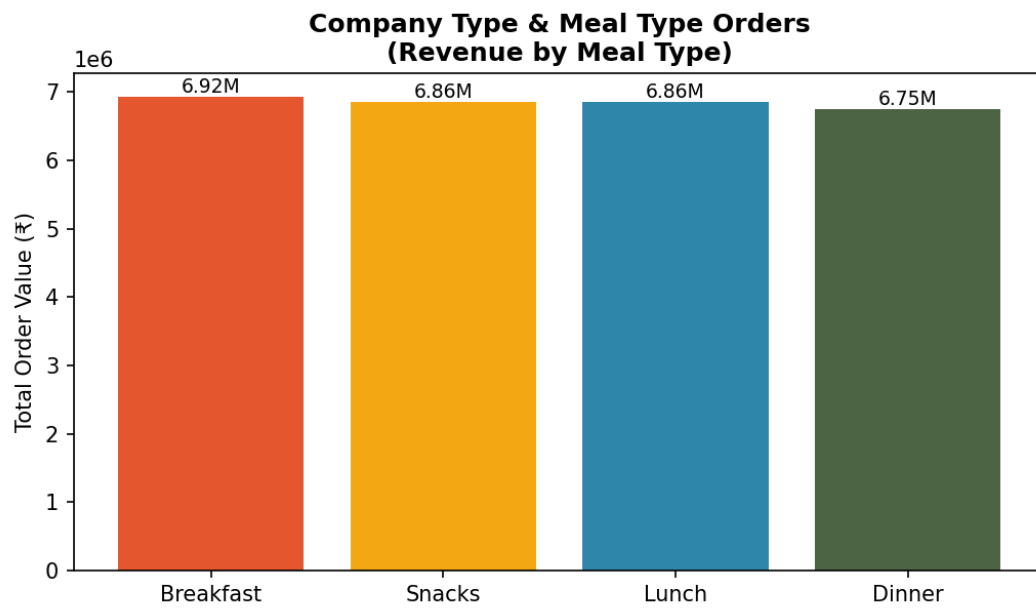


Figure 3. Total order value by meal type.

Story 1 — Guided Narrative

Story 1 sequences nine worksheets into a guided, presentation-ready narrative — starting from geographic and cuisine patterns, moving through demographic and revenue breakdowns, and ending on delivery-time performance and headline KPIs.

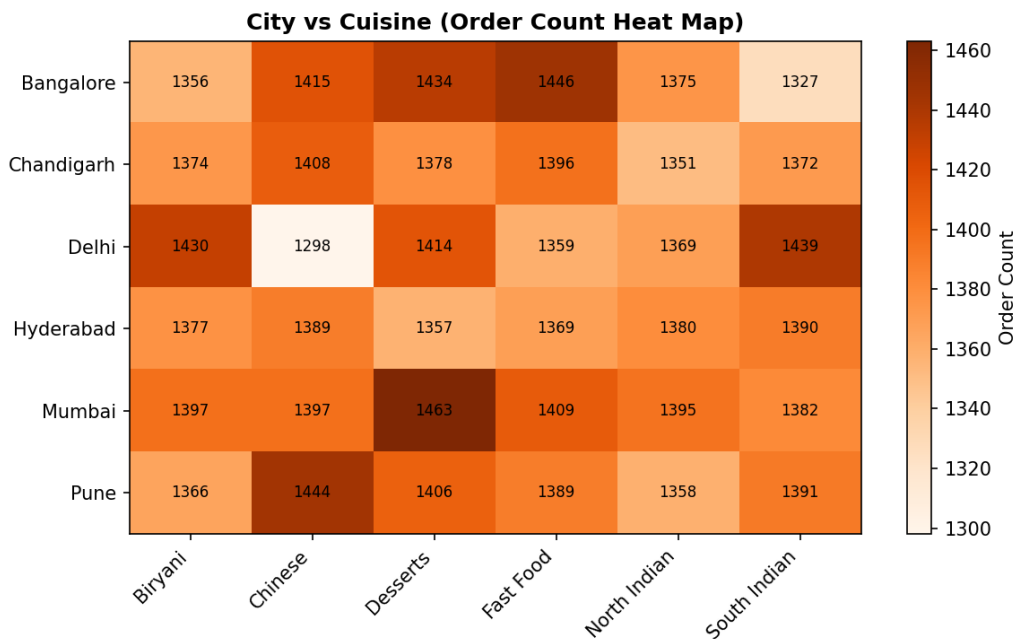


Figure 4. Order count by city and cuisine combination.

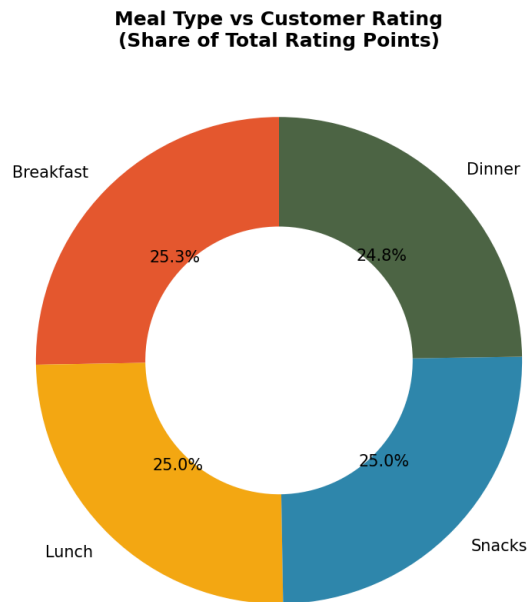


Figure 5. Share of total rating points contributed by each meal type.

Insight Delivery

Where parameter controls are visible, a viewer selects City, Cuisine, or Meal Type from the drop-down and watches every dependent worksheet update instantly — delivering a filtered insight in real time without writing a single query.

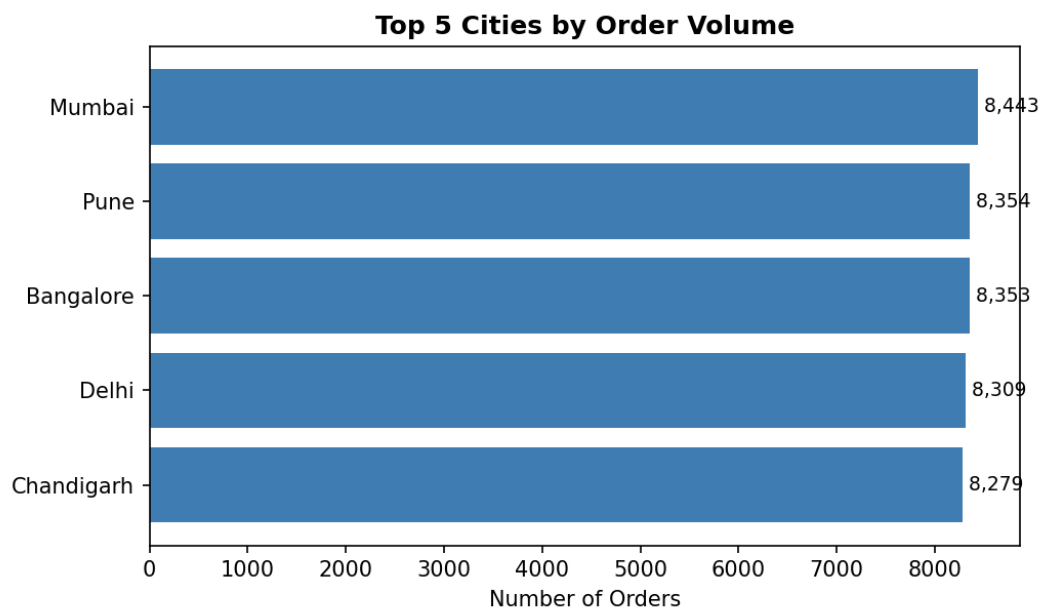


Figure 6. Top 5 cities ranked by number of orders placed.